

Project 02

AI Music & Audio Analyzer

Living Test & Development Report — v0.5

Item	Current Value
Project Status	In Progress — Audio Capture, Signal, Spectral & Rhythm Analysis Verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Input	Airdopes Joy Hands-Free microphone
Output	Airdopes Joy Stereo
Libraries	sounddevice + Librosa + NumPy + Matplotlib
Test Audio	5-second mono WAV — audio\airdopes_test.wav

1. Objective

Build a practical Python AI Music & Audio Analyzer using a Windows laptop and Airdopes Joy Bluetooth earbuds. The current phase validates hardware audio capture and progressively adds digital signal processing, visualization, spectral analysis, frequency-band analysis, and rhythm analysis.

2. Current Pipeline

Airdopes Joy Microphone → Bluetooth → Windows Audio → Python sounddevice → WAV → Librosa/NumPy → Waveform → Audio Level → FFT → Spectrogram → Spectral Features → Bass/Mid/Treble → Tempo/Beat Analysis

3. Completed Steps

Step	Test / Feature	Status
1	Audio device detection	PASS
2	Airdopes microphone verification	PASS
3	5-second WAV recording	PASS
4	WAV loading and basic analysis	PASS
5	Waveform visualization	PASS
6	Audio level analysis	PASS
7	FFT / frequency spectrum	PASS
8	STFT / spectrogram	PASS
9	Spectral features	PASS
10	Bass / Mid / Treble energy	PASS
11	Tempo / Beat analysis	PASS

4. Core Recording Results

Parameter	Actual Result
Airdopes Device ID	2
Input Channels	1
Sample Rate	44,100 Hz
Samples	220,500
Duration	5.00 sec
RMS Amplitude	0.031082
Peak Amplitude	0.423889
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS

5. Audio Level Analysis

Measurement	Result
Minimum Amplitude	-0.347687
Maximum Amplitude	0.423889
Mean Absolute Amplitude	0.010442
RMS Amplitude	0.031082
Peak Amplitude	0.423889
Crest Factor	13.638
Level Classification	Healthy
Output	reports\airdopes_audio_level.png

6. FFT / Frequency Spectrum

Measurement	Result
FFT Output Samples	110,251
Frequency Resolution	0.2000 Hz
Nyquist Frequency	22,050 Hz
Dominant Frequency	544.60 Hz
Dominant Magnitude	0.001539
Output	reports\airdopes_frequency_spectrum.png

The 544.60 Hz component is the strongest detected frequency component in the microphone-test recording. It is not treated as a definitive musical pitch.

7. STFT / Spectrogram

Measurement	Result
FFT Size	2048
Hop Length	512
Frequency Bins	1025
Time Frames	431
Frequency Resolution	21.53 Hz
Maximum Frequency	22,050 Hz
Output	reports\airdopes_spectrogram.png

8. Spectral Features

Feature	Actual Result
Spectral Centroid	7,159.79 Hz
Spectral Bandwidth	4,349.76 Hz
Spectral Rolloff (85%)	12,150.42 Hz
Spectral Flatness	0.329063
Zero Crossing Rate	0.147712
Brightness	Bright
Texture	Noise-like
Output	reports\airdopes_spectral_features.png

These classifications describe the current short microphone test signal and are not final music classifications.

9. Bass / Mid / Treble Analysis

Band	Range	Energy	Distribution
Bass	20–250 Hz	0.157129	3.49%
Mid	250–4,000 Hz	4.346717	96.48%
Treble	4,000–20,000 Hz	0.001499	0.03%
Dominant Band	—	—	Mid

The percentages are relative energy within the configured bands, not calibrated measurements of Airdopes frequency response.

10. Tempo / Beat Analysis — Step 11

Measurement	Actual Result
Estimated Tempo	147.66 BPM
Detected Beats	6
Detected Onsets	15
Average Onset Strength	0.498537
Maximum Onset Strength	7.015738
Tempo Category	Fast
Output	reports\airdopes_tempo_beats.png

Beat Timeline: Beat 01 — 1.045 sec | Beat 02 — 1.358 sec | Beat 03 — 1.718 sec | Beat 04 — 2.101 sec | Beat 05 — 2.485 sec | Beat 06 — 2.891 sec

The 147.66 BPM value is an algorithmic estimate for the 5-second microphone-test signal. It should not be considered the true musical tempo because the source is not a controlled music track.

11. Generated Analysis Files

File	Purpose	Status
audio\airdopes_test.wav	Raw Airdopes microphone recording	Generated
reports\airdopes_waveform.png	Time-domain waveform	Generated
reports\airdopes_audio_level.png	RMS level over time	Generated
reports\airdopes_frequency_spectrum.png	FFT frequency spectrum	Generated
reports\airdopes_spectrogram.png	STFT time-frequency view	Generated
reports\airdopes_spectral_features.png	Spectral feature visualization	Generated
reports\airdopes_bass_mid_treble.png	Frequency-band energy	Generated
reports\airdopes_tempo_beats.png	Onset and beat visualization	Generated

12. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

- audio/ → airdopes_test.wav
- reports/ → waveform, level, spectrum, spectrogram, spectral features, bass/mid/treble and tempo/beat graphs
- src/ → audio_devices.py, test_airdopes_mic.py, record_airdopes.py, analyze_airdopes_audio.py, plot_airdopes_waveform.py, audio_level_analysis.py, fft_frequency_analysis.py, spectrogram_analysis.py, spectral_features.py, bass_mid_treble.py, tempo_beat_analysis.py
- tests/
- venv/
- requirements.txt
- README.md

13. Overall Test Status

Validation	Status
Bluetooth Airdopes microphone accessible	PASS
Audio recording generated	PASS
WAV file reloadable	PASS
Time-domain waveform	PASS
RMS / Peak / dBFS	PASS
FFT spectrum	PASS
STFT spectrogram	PASS
Spectral feature extraction	PASS
Bass/Mid/Treble analysis	PASS
Tempo/Beat analysis	PASS
Integrated music-track validation	PENDING

14. Next Development Step

Step 12 — Integrated Audio Feature Summary. Combine outputs from Steps 4–11 into one structured analyzer and summary report. The next phase should also introduce a real music-track test so BPM, spectral balance and music-oriented features can be validated on an actual song.

15. Version History

- v0.1 — New Project-02 documentation baseline created from the beginning.
- v0.2 — Steps 4–6 documented: WAV analysis, waveform visualization and audio-level analysis.
- v0.3 — Steps 7–8 documented: FFT frequency spectrum and STFT spectrogram.
- v0.4 — Step 9 documented: spectral centroid, bandwidth, rolloff, flatness and zero-crossing rate.
- v0.5 — Steps 10–11 documented: Bass/Mid/Treble energy analysis and Tempo/Beat analysis with actual results and generated visualizations.

Only the new Project-02 history is included; previous report versions are intentionally excluded.